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NEET · CHEMISTRY · CHAPTER-WISE PYQS

Chemical Bonding and Molecular Structure

NEET Chemistry · Previous Year Questions with Answer Key

168	50	1988 – 2026	English
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MCQ (Single Correct Answer)	166 questions
MCQ (More than One Correct Answer)	2 questions
Answer key	at the end

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MCQ (Single Correct Answer) 166 questions

Q1. RE-NEET 2026

The highest occupied molecular orbital for Ne_2 is

(A) σ^*_{2p} (B) π_{2p} (C) σ_{2p} (D) π^*_{2p}

Open solution

Q2. RE-NEET 2026

Match the species in List-I with their geometry in List-II.

List-I	List-II
A. PCl_5	I. Tetrahedral
B. BrF_5	II. Square Planar
C. BF_4^-	III. Trigonal bipyramidal
D. $[Ni(CN)_4]^{2-}$	IV. Square pyramidal

Choose the correct answer from the options given below:

(A) A-III, B-II, C-I, D-IV (B) A-IV, B-III, C-I, D-II

(C) A-III, B-IV, C-I, D-II (D) A-III, B-I, C-II, D-IV

Open solution

Q3. NEET 2026

Match List I with List II :

List-I		List-II	
A.	C_2H_4	I.	3σ bonds, 2π bonds
B.	C_2H_2	II.	3σ bonds, one lone pair
C.	CH_4	III.	4σ bonds
D.	NH_3	IV.	5σ bonds, 1π bond

Choose the correct answer from the options given below :

(A) A-III, B-IV, C-II, D-I (B) A-IV, B-I, C-III, D-II

(C) A-I, B-II, C-IV, D-III (D) A-II, B-III, C-I, D-IV

Open solution

Q4. NEET 2026

Identify the correct statement about ClF_3 from the following options :

(A) It has T-shaped geometry with two lone pairs on Cl atom.

(B) It has T-shaped geometry with three lone pairs on Cl atom.

(C) It has a trigonal pyramidal geometry with two lone pairs on Cl atom.

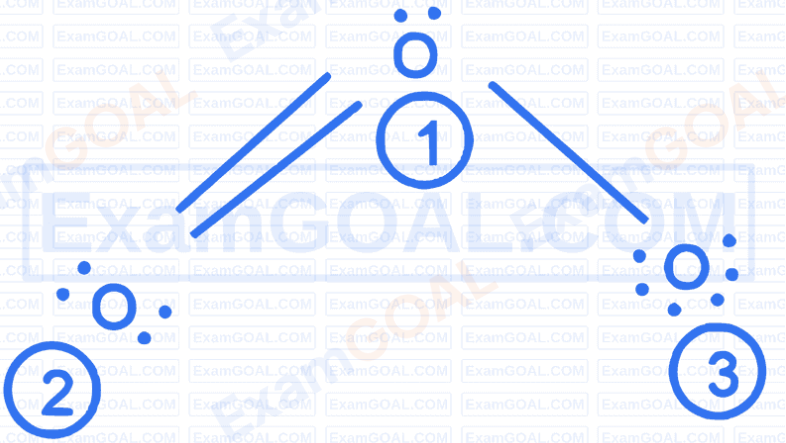
(D) It has a planar trigonal geometry with two lone pairs on Cl atom.

Open solution

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Q5. NEET 2026

The correct formal charges on oxygen atoms numbered $2, 1$ and 3 respectively are :



Open solution

(A) $-1, 0, +1$ (B) $0, +1, -1$ (C) $0, 0, 0$ (D) $+1, 0, -1$

Q6. NEET 2025

Given below are two statements :

Statement I: A hypothetical diatomic molecule with bond order zero is quite stable.

Statement II: As bond order increases, the bond length increases.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Open solution

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Q7. NEET 2025

Open solution

Match List-I with List-II

	List - I		List - II
A	XeO_3	I	sp^3 d; linear
B	XeF_2	II	sp^3; pyramidal
C	$XeOF_4$	III	sp^3d^3; distorted octahedral
D	XeF_6	IV	sp^3d^2; square pyramidal

Choose the correct answer from the options given below :

(A) A-IV, B-II, C-III, D-I
(B) A-IV, B-II, C-I, D-III
(C) A-II, B-I, C-IV, D-III
(D) A-II, B-I, C-III, D-IV

Q8. NEET 2025

Open solution

Identify the correct orders against the property mentioned

A. $H_2O > NH_3 > CHCl_3$-dipole moment

B. $XeF_4 > XeO_3 > XeF_2$ - number of lone pairs on central atom

C. $O - H > C - H > N - O$– bond length

D. $N_2 > O_2 > H_2$– bond enthalpy

Choose the correct answer from the options given below:

(A) A, C only
(B) B, C only
(C) A, D only
(D) B, D only

Q9. NEET 2024 (Re-Examination)

Open solution

Which of the following molecules has "NON ZERO" dipole moment value?

(A) CCl_4
(B) HI
(C) CO_2
(D) BF_3

Q10. NEET 2024 (Re-Examination)

Open solution

Arrange the following compounds in increasing order of their solubilities in chloroform:
$NaCl$, CH_3OH, cyclohexane, CH_3CN

(A) $NaCl < CH_3CN < CH_3OH <$ Cyclohexane
(B) $CH_3OH < CH_3CN < NaCl <$ Cyclohexane
(C) $NaCl < CH_3OH < CH_3CN <$ Cyclohexane
(D) Cyclohexane $< CH_3CN < CH_3OH < NaCl$

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Q11. NEET 2024 (Re-Examination)

Open solution

Identify the incorrect statement about PCl_5.

(A) PCl_5 possesses two different $Cl - P - Cl$ bond angles

(B) All five $P - Cl$ bonds are identical in length

(C) PCl_5 exhibits sp^3d hybridisation

(D) PCl_5 consists of five $P - Cl$ (sigma) bonds

Q12. NEET 2024 (Re-Examination)

Open solution

Match List-I with List-II:

	List - I Molecule		**List - II Bond enthalpy (kJ mol^{-1})**
(A)	HCl	(I)	435.8
(B)	N_2	(II)	498
(C)	H_2	(III)	946.0
(D)	O_2	(IV)	431.0

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II

(B) A-IV, B-I, C-III, D-II

(C) A-IV, B-III, C-II, D-I

(D) A-IV, B-III, C-I, D-II

Q13. NEET 2024

Open solution

Match List I with List II.

	List I (Molecule)		**List II (Number and types of bond/s between two carbon atoms)**
A.	ethane	I.	one σ-bond and two π-bonds
B.	ethene	II.	two π-bonds
C.	carbon molecule, C_2	III.	one σ-bond
D.	ethyne	IV.	one σ-bond and one π-bond

Choose the correct answer from the options given below :

(A) A-I, B-IV, C-II, D-III

(B) A-IV, B-III, C-II, D-I

(C) A-III, B-IV, C-II, D-I

(D) A-III, B-IV, C-I, D-II

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Q14. NEET 2024

Open solution

Given below are two statements:

Statement I: The boiling point of hydrides of Group 16 elements follow the order $H_2O > H_2Te > H_2Se > H_2S$.

Statement II: On the basis of molecular mass, H_2O is expected to have lower boiling point than the other members of the group but due to the presence of extensive H-bonding in H_2O, it has higher boiling point.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

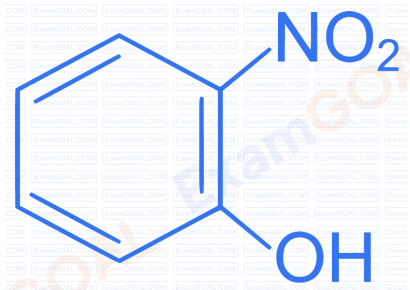
(C) Statement I is true but Statement II is false

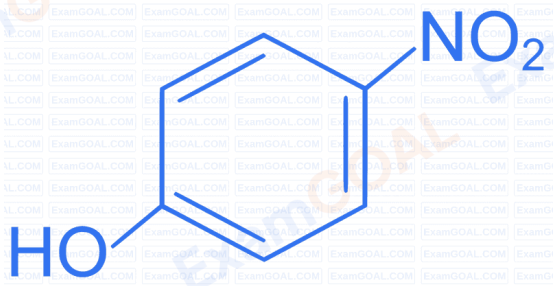
(D) Statement I is false but Statement II is true

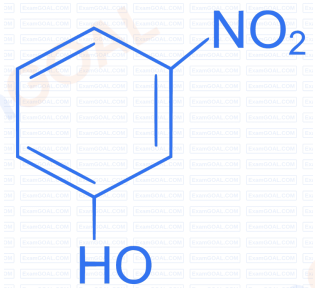
Q15. NEET 2024

Open solution

Intramolecular hydrogen bonding is present in

(A) 

(B) 

(C) 

(D) HF

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Q16. NEET 2024

Match List I with List II.

	List I (Compound)		List II (Shape/geometry)
A.	NH_3	I.	Trigonal Pyramidal
B.	BrF_5	II.	Square Planar
C.	XeF_4	III.	Octahedral
D.	SF_6	IV.	Square Pyramidal

Choose the correct answer from the options given below:

(A) A-I, B-IV, C-II, D-III

(B) A-II, B-IV, C-III, D-I

(C) A-III, B-IV, C-I, D-II

(D) A-II, B-III, C-IV, D-I

Open solution

Q17. NEET 2024

Identify the correct answer.

(A) Three resonance structures can be drawn for ozone

(B) BF_3 has non-zero dipole moment

(C) Dipole moment of NF_3 is greater than that of NH_3

(D) Three canonical forms can be drawn for CO_3^{2-} ion

Open solution

Q18. NEET 2023 Manipur

The correct order of dipole moments for molecules NH_3, H_2S, CH_4 and HF, is:

(A) $CH_4 > H_2S > NH_3 > HF$

(B) $H_2S > NH_3 > HF > CH_4$

(C) $NH_3 > HF > CH_4 > H_2S$

(D) $HF > NH_3 > H_2S > CH_4$

Open solution

Q19. NEET 2023 Manipur

Which one of the following represents all isoelectronic species ?

(A) Na^+, Cl^-, O^-, NO^+

(B) N_2O, N_2O_4, NO^+, NO

(C) Na^+, Mg^{2+}, O^-, F^-

(D) Ca^{2+}, Ar, K^+, Cl^-

Open solution

Q20. NEET 2023 Manipur

Which one of the following statements is incorrect related to Molecular Orbital Theory?

(A) The π^* antibonding molecular orbital has a node between the nuclei.

(B) In the formation of bonding molecular orbital, the two electron waves of the bonding atoms reinforce each other.

(C) Molecular orbitals obtained from $2P_x$ and $2P_y$ orbitals are symmetrical around the bond axis.

(D) A π-bonding molecular orbital has larger electron density above and below the internuclear axis.

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Q21. NEET 2023 Manipur

Open solution

Given below are two statements:

Statement I : Hydrated chlorides and bromides of Ca, Sr and Ba on heating undergo hydrolysis.

Statement II : Hydrated chlorides and bromides of Be and Mg on heating undergo dehydration.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement-I is correct but Statement-II is false.

(B) Statement-I is incorrect but Statement-II is true.

(C) Both Statement-I and Statement-II are true.

(D) Both Statement-I and Statement-II are false.

Q22. NEET 2023

Open solution

Amongst the following, the total number of species NOT having eight electrons around central atom in its outer most shell, is

$NH_3, AlCl_3, BeCl_2, CCl_4, PCl_5$:

(A) 2 (B) 4 (C) 1 (D) 3

Q23. NEET 2023

Open solution

The correct order of energies of molecular orbitals of N_2 molecule, is

(A) $\sigma 1\,s < \sigma^* 1\,s < \sigma 2\,s < \sigma^* 2\,s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

(B) $\sigma 1\,s < \sigma^* 1\,s < \sigma 2\,s < \sigma^* 2\,s < \sigma 2p_z < \sigma^* 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y)$

(C) $\sigma 1\,s < \sigma^* 1\,s < \sigma 2\,s < \sigma^* 2\,s < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma 2p_z < \sigma^* 2p_z$

(D) $\sigma 1\,s < \sigma^* 1\,s < \sigma 2\,s < \sigma^* 2\,s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

Q24. NEET 2023

Open solution

Intermolecular forces are forces of attraction and repulsion between interacting particles that will include:

A. dipole - dipole forces.

B. dipole - induced dipole forces

C. hydrogen bonding

D. covalent bonding

E. dispersion forces

Choose the most appropriate answer from the options given below :

(A) A, B, C, D are correct

(B) A, B, C, E are correct

(C) A, C, D, E are correct

(D) B, C, D, E are correct

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Q25. NEET 2022 Phase 2

Open solution

Match List-I with List-II :

	List-I (Molecules)		List-II (Shape)
(a)	NH_3	(i)	Square pyramidal
(b)	ClF_3	(ii)	Trigonal bipyramidal
(c)	PCl_5	(iii)	Trigonal pyramidal
(d)	BrF_5	(iv)	T-shape

Choose the correct answer from the options given below :

(A) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

(B) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

(C) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

(D) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

Q26. NEET 2022 Phase 2

Open solution

The correct order of bond angles in the following compounds/species is :

(A) $CO_2 < NH_3 < H_2O < \overset{+}{N}H_4$

(B) $H_2O < NH_3 < \overset{+}{N}H_4 < CO_2$

(C) $H_2O < \overset{+}{N}H_4 < NH_3 < CO_2$

(D) $H_2O < \overset{+}{N}H_4 = NH_3 < CO_2$

Q27. NEET 2022 Phase 1

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : ICl is more reactive than I_2

Reason (R) : I-Cl bond is weaker than I-I bond.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(C) (A) is correct but (R) is not correct.

(D) (A) is not correct but (R) is correct.

Q28. NEET 2022 Phase 1

Open solution

Amongst the following which one will have maximum 'lone pair - lone pair' electron repulsions?

(A) ClF_3 **(B)** IF_5 **(C)** SF_4 **(D)** XeF_2

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Q29. NEET 2022 Phase 1

Which amongst the following is incorrect statement?

(A) The bond orders of O_2^+, O_2, O_2^- and O_2^{2-} are 2.5, 2, 1.5 and 1, respectively

(B) C_2 molecule has four electrons in its two degenerate π molecular orbitals

(C) H_2^+ ion has one electron

(D) O_2^+ ion is diamagnetic

Open solution

Q30. NEET 2021

BF_3 is planar and electron deficient compound. Hybridization and number of electrons around the central atom, respectively are :

(A) sp^2 and 8 (B) sp^3 and 4 (C) sp^3 and 6 (D) sp^2 and 6

Open solution

Q31. NEET 2021

Match List - I with List - II :

	List - I		List - II
(a)	PCl_5	(i)	Square pyramidal
(b)	SF_6	(ii)	Trigonal planar
(c)	BrF_5	(iii)	Octahedral
(d)	BF_3	(iv)	Trigonal bipyramidal

Choose the correct answer from the options given below.

(A) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i) (B) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(C) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i) (D) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Open solution

Q32. NEET 2021

The correct sequence of bond enthalpy of 'C-X' bond is :

(A) CH_3 - Cl > CH_3 - F > CH_3 - Br > CH_3 - I (B) CH_3 - F < CH_3 - Cl < CH_3 - Br < CH_3 - I

(C) CH_3 - F > CH_3 - Cl > CH_3 - Br > CH_3 - I (D) CH_3 - F < CH_3 - Cl > CH_3 - Br > CH_3 - I

Open solution

Q33. NEET 2021

Which of the following molecules is non-polar in nature?

(A) NO_2 (B) $POCl_3$ (C) CH_2O (D) $SbCl_5$

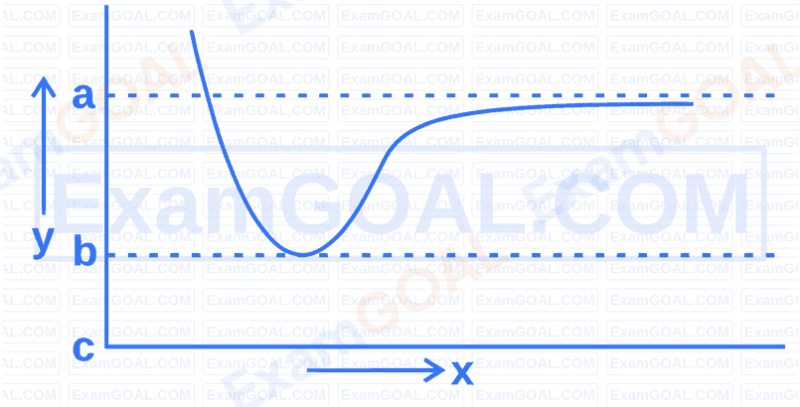
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Q34. NEET 2020 Phase 2

The potential energy (y) curve for H_2 formation as a function of internuclear distance (x) of the H atoms is shown below.

Open solution



The bond energy of H_2 is

(A) $(c-a)$ (B) $(b-a)$ (C) $\frac{(c-a)}{2}$ (D) $\frac{(b-a)}{2}$

Q35. NEET 2020 Phase 2

Among the compounds shown below which one revealed a linear structure?

(A) N_2O (B) NO_2 (C) $HOCl$ (D) O_3

Open solution

Q36. NEET 2020 Phase 2

Identify the wrongly matched pair.

Molecule – Shape or geometry of molecule

(A) NH_3 – Trigonal pyramidal (B) PCl_5 – Trigonal planar

(C) SF_6 – Octahedral (D) $BeCl_2$ – Linear

Open solution

Q37. NEET 2020 Phase 2

How many (i) sp^2 hybridised carbon atoms and (ii) π bonds are present in the following compound?



(A) 8, 5 (B) 7, 5 (C) 8, 6 (D) 7, 6

Open solution

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Q38. NEET 2020 Phase 1

Which of the following set of molecules will have zero dipole moment?

(A) Boron trifluoride, hydrogen fluoride, carbon dioxide, 1, 3-dichlorobenzene

(B) Nitrogen trifluoride, beryllium difluoride, water, 1, 3-dichlorobenzene

(C) Boron trifluoride, beryllium difluoride, carbon dioxide, 1, 4-dichlorobenzene

(D) Ammonia, beryllium difluoride, water, 1, 4-dichlorobenzene

Open solution

Q39. NEET 2020 Phase 1

Identify a molecule which does not exist.

(A) Li_2 (B) C_2 (C) O_2 (D) He_2

Open solution

Q40. NEET 2019 (Odisha)

Which of the following is paramagnetic?

(A) O_2 (B) N_2 (C) H_2 (D) Li_2

Open solution

Q41. NEET 2019 (Odisha)

Which of the following is the correct order of dipole moment?

(A) $H_2O < NF_3 < NH_3 < BF_3$ (B) $NH_3 < BF_3 < NF_3 < H_2O$

(C) $BF_3 < NF_3 < NH_3 < H_2O$ (D) $BF_3 < NH_3 < NF_3 < H_2O$

Open solution

Q42. NEET 2019

The number of sigma (σ) and pi (π) bonds in pent-2-en-4-yne is -

(A) 11 σ bonds and 2 π bonds (B) 13 σ bonds and no π bonds

(C) 10 σ bonds and 3 π bonds (D) 8 σ bonds and 5 π bonds

Open solution

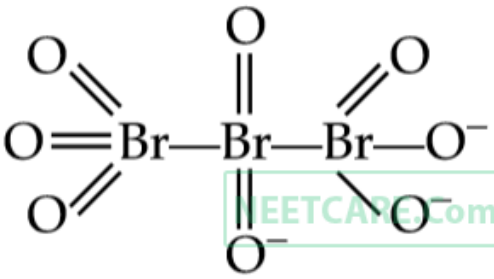
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Q43. NEET 2019

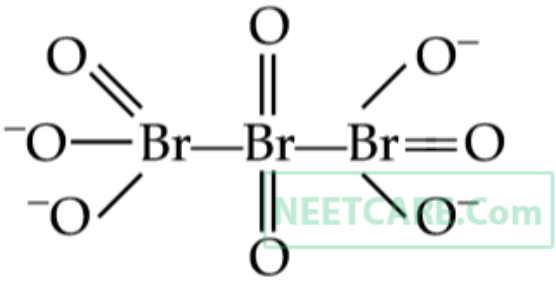
The correct structure of tribromooctaoxide is :

Open solution

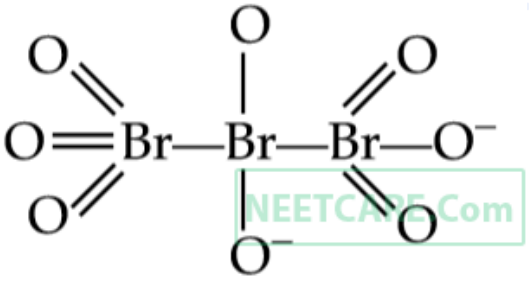
(A)



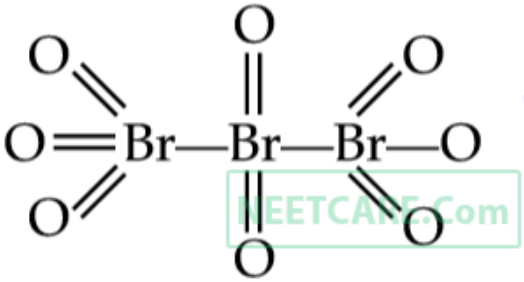
(B)



(C)



(D)



Q44. NEET 2019

The manganate and permanganate ions are tetrahedral due to :

Open solution

(A) The π-bonding involves overlap of p-orbitals of oxygen with p-orbtials of manganese

(B) The π-bonding involves overlap of d-orbitals of oxygen with d-orbitals of manganese

(C) The π-bonding involves overlap of p-orbitals of oxygen with d-orbitals of manganese

(D) There is no π-bonding

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Q45. NEET 2019

Which of the following diatomic molecular species has only π bonds according to Molecular Orbital Theory?

(A) C_2 (B) Be_2 (C) O_2 (D) N_2

Open solution

Q46. NEET 2018

Consider the following species : CN^+ , CN^- , NO and CN. Which one of these will have the highest bond order?

(A) NO (B) CN^- (C) CN^+ (D) CN

Open solution

Q47. NEET 2018

In the structure of ClF_3 , the number of lone pairs of electrons on central atom 'Cl' is

(A) one (B) two (C) four (D) three

Open solution

Q48. NEET 2017

The species, having bond angles of 120° is

(A) ClF_3 (B) NCl_3 (C) BCl_3 (D) PH_3

Open solution

Q49. NEET 2017

Which one of the following pairs of species have the same bond order ?

(A) O_2, NO^+ (B) CN^-, CO (C) N_2, O_2^- (D) CO, NO

Open solution

Q50. NEET 2016 Phase 2

Which one of the following compounds shows the presence of intramolecular hydrogen bond?

(A) H_2O_2 (B) HCN

(C) Cellulose (D) Concentrated acetic acid

Open solution

Q51. NEET 2016 Phase 2

The hybridizations of atomic orbitals of nitrogen in NO_2^+, NO_3^- and NH_4^+ respectively are

(A) sp, sp^3 and sp^2 (B) sp^2, sp^3 and sp^3 (C) sp, sp^2 and sp^3 (D) sp^2, sp and sp^3

Open solution

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Q52. NEET 2016 Phase 2

The correct geometry and hybridization for XeF_4 are

(A) octahedral, sp^3d^2

(B) trigonal bipyramidal, sp^3d

(C) planar triangle, sp^3d^3

(D) square planar, sp^3d^2

Open solution

Q53. NEET 2016 Phase 2

Among the following, which one is a wrong statement ?

(A) PH_5 and $BiCl_5$ do not exist.

(B) $p\pi$-$d\pi$ bonds are present in SO_2.

(C) SeF_4 and CH_4 have same shape.

(D) I_3^+ has bent geometry.

Open solution

Q54. NEET 2016 Phase 1

Consider the molecules CH_4, NH_3 and H_2O.

Which of the given statements is false?

(A) The H – O – H bond angle in H_2O is smaller than the H – N – H bond angle in NH_3.

(B) The H – C – H bond angle in CH_4 is larger than the H – N – H bond angle in NH_3.

(C) The H – C – H bond angle in CH_4, the H – N – H bond angle in NH_3, and the H – O – H bond angle in H_2O are all greater than 90°.

(D) The H – O – H bond angle in H_2O is larger than the H – C – H bond angle in CH_4.

Open solution

Q55. NEET 2016 Phase 1

Predict the correct order among the following :

(A) bond pair - bond pair > lone pair - bond pair > lone pair - lone pair

(B) lone pair - bond pair > bond pair - bond pair > lone pair - lone pair

(C) lone pair - lone pair > lone pair - bond pair > bond pair - bond pair

(D) lone pair - lone pair > bond pair - bond pair > lone pair - bond pair

Open solution

Q56. AIPMT 2015

In which of the following pairs, both the species are not isostructural ?

(A) Diamond, Silicon carbide

(B) NH_3, PH_3

(C) XeF_4, XeO_4

(D) $SiCl_4$, PCl_4^+

Open solution

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Q57. AIPMT 2015

Decreasing order of stability of O_2, O_2^-, O_2^+ and O_2^{2-} is

(A) $O_2^{2-} > O_2^- > O_2 > O_2^+$

(B) $O_2 > O_2^+ > O_2^{2-} > O_2^-$

(C) $O_2^- > O_2^{2-} > O_2^+ > O_2$

(D) $O_2^+ > O_2 > O_2^- > O_2^{2-}$

Open solution

Q58. AIPMT 2015 Cancelled Paper

Which of the following pairs of ions are isoelectronic and isostructural ?

(A) SO_3^{2-}, NO_3^-

(B) ClO_3^-, SO_3^{2-}

(C) CO_3^{2-}, SO_3^{2-}

(D) ClO_3^-, CO_3^{2-},

Open solution

Q59. AIPMT 2015 Cancelled Paper

The correct bond order in the following species is

(A) $O_2^+ < O_2^- < O_2^{2+}$

(B) $O_2^- < O_2^+ < O_2^{2+}$

(C) $O_2^{2+} < O_2^+ < O_2^-$

(D) $O_2^{2+} < O_2^- < O_2^+$

Open solution

Q60. AIPMT 2015 Cancelled Paper

Which of the following options represents the correct bond order ?

(A) $O_2^- > O_2 < O_2^+$

(B) $O_2^- < O_2 > O_2^+$

(C) $O_2^- > O_2 > O_2^+$

(D) $O_2^- < O_2 < O_2^+$

Open solution

Q61. AIPMT 2015 Cancelled Paper

Maximum bond angle at nitrogen is present in which of the following ?

(A) NO_2^+

(B) NO_3^-

(C) NO_2

(D) NO_2^-

Open solution

Q62. AIPMT 2015 Cancelled Paper

Which of the following species contains equal number of σ- and π-bonds ?

(A) $(CN)_2$

(B) $CH_2(CN)_2$

(C) HCO_3^-

(D) XeO_4

Open solution

Q63. AIPMT 2014

Which of the following molecules has the maximum dipole moment ?

(A) CO_2

(B) CH_4

(C) NH_3

(D) NF_3

Open solution

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Q64. AIPMT 2014

Which one of the following species has plane triangular shape ?

(A) N_3^- (B) NO_3^- (C) NO_2^- (D) CO_2

Open solution

Q65. NEET 2013 (Karnataka)

The pair of species that has the same bond order in the following is

(A) CO, NO^+ (B) NO^-, CN^- (C) O_2, N_2 (D) O_2, B_2

Open solution

Q66. NEET 2013 (Karnataka)

The outer orbitals of C in ethene molecule can be considered to be hybridized to give three equivalent sp^2 orbitals. The total number of sigma (σ) and pi (π) bonds in ethene molecule is

(A) 3 sigma (σ) and 2 pi (π) bonds

(B) 4 sigma (σ) and 1 pi (π) bonds

(C) 5 sigma (σ) and 1 pi (π) bonds

(D) 1 sigma (σ) and 2 pi (π) bonds

Open solution

Q67. NEET 2013 (Karnataka)

In which of the following pair both the species have sp^3 hybridization ?

(A) SiF_4, BeH_2 (B) NF_3, H_2O (C) NF_3, BF_3 (D) H_2S, BF_3

Open solution

Q68. NEET 2013 (Karnataka)

In which of the following ionization processes the bond energy increases and the magnetic behaviour changes from paramagnetic to diamagnetic.

(A) $O_2 \rightarrow O_2^+$ (B) $C_2 \rightarrow C_2^+$ (C) $NO \rightarrow NO^+$ (D) $N_2 \rightarrow N_2^+$

Open solution

Q69. NEET 2013

Which of the following is electron-deficient ?

(A) $(BH_3)_2$ (B) PH_3 (C) $(CH_3)_2$ (D) $(SiH_3)_2$

Open solution

Q70. NEET 2013

XeF_2 is isostructural with

(A) $SbCl_3$ (B) $BaCl_2$ (C) TeF_2 (D) ICl_2^-

Open solution

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Q71. NEET 2013

Which of the following is a polar molecule?

(A) SiF_4 (B) XeF_4 (C) BF_3 (D) SF_4

Open solution

Q72. NEET 2013

Which of the following is paramagnetic?

(A) CN^- (B) NO^+ (C) CO (D) O_2^-

Open solution

Q73. NEET 2013

Dipole-induced dipole interactions are present in which of the following pairs

(A) HCl and He atoms (B) SiF_4 and He atoms

(C) H_2O and alcohol (D) Cl_2 and CCl_4

Open solution

Q74. NEET 2013

Identify the correct order of solubility in aqueous medium.

(A) Na_2S > CuS > ZnS (B) Na_2S > ZnS > CuS (C) CuS > ZnS > Na_2S (D) ZnS > Na_2S > CuS

Open solution

Q75. AIPMT 2012 Mains

During change of O_2 to O_2^- ion, the electron adds on which one of the following orbitals ?

(A) π^* orbital (B) π orbital (C) σ^* orbital (D) σ orbital

Open solution

Q76. AIPMT 2012 Mains

Four diatomic species are listed below. Identify the correct order in which the bond order is increasing in them

(A) $NO < {O_2}^- < {C_2}^{2-} < He_2^+$ (B) ${O_2}^- < NO < {C_2}^{2-} < He_2^+$

(C) ${C_2}^{2-} < He_2^+ < {O_2}^- < NO$ (D) $He_2^+ < {O_2}^- < NO < {C_2}^{2-}$

Open solution

Q77. AIPMT 2012 Prelims

Which one of the following pairs is isostructural (i.e., having the same shape and hybridization) ?

(A) [BCl_3 and $BrCl_3$] (B) [NH_3 and NO_3^-]

(C) [NF_3 and BF_3] (D) [BF_4^- and NH_4^+]

Open solution

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Q78. AIPMT 2012 Prelims

Bond order of 1.5 is shown by

(A) O_2^+ (B) O_2^- (C) O_22^- (D) O_2

Open solution

Q79. AIPMT 2012 Prelims

Which of the following species contains three bond pairs and one lone pair arround the central atom ?

(A) H_2O (B) BF_3 (C) NH_2^- (D) PCl_3

Open solution

Q80. AIPMT 2012 Prelims

The pair of species with the same bond order is

(A) O_2^{2-}, B_2 (B) O_2^+, NO^+ (C) NO, CO (D) N_2, O_2

Open solution

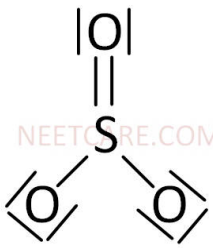
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Q81. AIPMT 2011 Mains

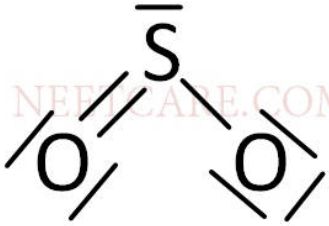
Which of the following structures is the most preferred and hence of lowest energy for SO_3?

Open solution

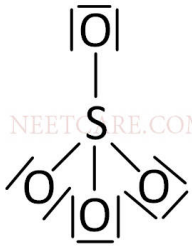
(A)



(B)



(C)



(D)



Q82. AIPMT 2011 Mains

The pairs of species of oxygen and their magnetic behaviours are noted below. Which of the following presents the correct description?

Open solution

(A) O_2^-, O_2^{2-} - Both diamagnetic

(B) O_2^+, O_2 - Both paramagnetic

(C) O, O_2^{2-} - Both paramagnetic

(D) O^+, O_2^{2-} - Both paramagnetic

Q83. AIPMT 2011 Prelims

Which of the following has the minimum bond length ?

Open solution

(A) O_2^+

(B) O_2^-

(C) O_2^{2-}

(D) O_2

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Q84. AIPMT 2011 Prelims

Which of the two ions from the list given below that have the geometry that is explained by the same hybridization of orbitals, NO_2^-, NO_3^-, NH_2^-, NH_4^+, SCN^-?

(A) NO_2^- and NO_3^-
(B) NH_4^+ and NO_3^-
(C) SCN^- and NH_2^-
(D) NO_2^- and NH_2^-

Open solution

Q85. AIPMT 2011 Prelims

The correct order of increasing bond length of C – H, C – O, C – C and C=C is

(A) C – H < C=C < C – O < C – C
(B) C – C < C=C < C – O < C – H
(C) C – O < C – H < C – C < C=C
(D) C – H < C – O < C – C < C=C

Open solution

Q86. AIPMT 2011 Prelims

The pairs of species of oxygen and their magnetic behavior are noted below. Which of the following presents the correct description?

(A) O_2^-, O_2^{2-} - Both diamagnetic
(B) O^+, O_2^{2-} - Both paramagnetic
(C) O_2^+, O_2 - Both paramagnetic
(D) O, O_2^{2-} - Both paramagnetic

Open solution

Q87. AIPMT 2010 Mains

In which of the following molecules the central atom does not have sp^3 hybridization?

(A) CH_4
(B) SF_4
(C) BF_4^-
(D) NH_4^+

Open solution

Q88. AIPMT 2010 Mains

Some of the properties of the two species, NO_3^- and H_3O^+ are described below. Which one of them is correct?

(A) Dissimilar in hybridization for the central atom with different structures.
(B) Isostructural with same hybridization for the central atom.
(C) Isostructural with different hybridization for the central atom
(D) Similar in hybridization for the central atom with different structures.

Open solution

Q89. AIPMT 2010 Prelims

In which of the following pairs of molecules/ ions, the central atoms have sp^2 hybridisation ?

(A) NO_2^- and NH_3
(B) BF_3 and NO_2^-
(C) NH_2^- and H_2O
(D) BF_3 and NH_2^-

Open solution

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Q90. AIPMT 2010 Prelims

Which one of the following species does not exist under normal conditions?

(A) Be_2^+ (B) Be_2 (C) B_2 (D) Li_2

Open solution

Q91. AIPMT 2010 Prelims

In which one of the following species the central atom has the type of hybridization which is not the same as that present in the other three ?

(A) SF_4 (B) I_3^- (C) $SbCl_5^{2-}$ (D) PCl_5

Open solution

Q92. AIPMT 2009

What is the dominant intermolecular force or bond that must be overcome in converting liquid CH_3OH to a gas?

(A) Dipole-dipole interaction (B) Covalent bonds

(C) London dispersion force (D) Hydrogen bonding

Open solution

Q93. AIPMT 2009

According to MO theory which of the lists ranks the nitrogen species in terms of increasing bond order ?

(A) $N_2^{2-} < N_2^- < N_2$ (B) $N_2 < N_2^{2-} < N_2^-$

(C) $N_2^- < N_2^{2-} < N_2$ (D) $N_2^- < N_2 < N_2^{2-}$

Open solution

Q94. AIPMT 2009

In which of the following molecules/ions BF_3, NO_2^-, NH_2^- and H_2O, the central atom is sp^2 hybridised?

(A) NH_2^- and H_2O (B) NO_2^- and H_2O

(C) BF_3 and NO_2^- (D) NO_2^- and NH_2^-

Open solution

Q95. AIPMT 2008

Four diatomic species are listed below. Identify the correct order in which the bond order is increasing in them

(A) $NO < {O_2}^- < {C_2}^{2-} < He_2^+$ (B) ${O_2}^- < NO < {C_2}^{2-} < He_2^+$

(C) ${C_2}^{2-} < He_2^+ < {O_2}^- < NO$ (D) $He_2^+ < {O_2}^- < NO < {C_2}^{2-}$

Open solution

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Q96. AIPMT 2008

The correct order of increasing bond angles in the following triatomic species is

(A) $NO_2^+ < NO_2 < NO_2^-$

(B) $NO_2^+ < NO_2^- < NO_2$

(C) $NO_2^- < NO_2^+ < NO_2$

(D) $NO_2^- < NO_2 < NO_2^+$

Open solution

Q97. AIPMT 2007

In which of the following pairs, the two species are isostructural?

(A) SO_3^{2-} and NO_3^-

(B) BF_3 and NF_3

(C) BrO_3^- and XeO_3

(D) SF_4 and XeF_4

Open solution

Q98. AIPMT 2007

The correct order of C – O bond length among CO, CO_3^{2-}, CO_2 is

(A) $CO < CO_3^{2-} < CO_2$

(B) $CO_3^{2-} < CO_2 < CO$

(C) $CO < CO_2 < CO_3^{2-}$

(D) $CO_2 < CO_3^{2-} < CO$

Open solution

Q99. AIPMT 2006

Which of the following is not a correct statement?

(A) Multiple bonds are always shorter than corresponding single bonds.

(B) The electron-deficient molecules can act as Lewis acids.

(C) The canonical structures have no real existence.

(D) Every AB_5 molecule does in fact have square pyramid structure.

Open solution

Q100. AIPMT 2006

Which of the following is not isostructural with $SiCl_4$?

(A) NH_4^+

(B) SCl_4

(C) SO_4^{2-}

(D) PO_4^{3-}

Open solution

Q101. AIPMT 2006

Which of the following species has a linear shape?

(A) O_3

(B) NO_2^-

(C) SO_2

(D) NO_2^+

Open solution

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Q102. AIPMT 2006

The electronegativity difference berween N and F is greater than that between N and H yet the dipole moment of NH_3 (1.5D) is larger than that of NF_3 (0.2 D). This is because

(A) in NH_3 the atomic dipole and bond dipole are in the opposite directions whereas in NF_3 these are in the same direction

(B) in NH_3 as well as in NF_3 the atomic dipole and bond dipole are in the same direction

(C) in NH_3 the atomic dipole and bond dipole are in the same direction whereas in NF_3 these are in opposite directions

(D) in NH_3 as well as in NF_3 the atomic dipole and bond dipole are in opposite directions.

Open solution

Q103. AIPMT 2006

The number of unpaired electrons in a paramagnetic diatomic molecule of an element with atomic number 16 is

(A) 2 (B) 3 (C) 4 (D) 1

Open solution

Q104. AIPMT 2005

Which of the following molecules has trigonal planar geometry?

(A) BF_3 (B) NH_3 (C) PCl_3 (D) IF_3

Open solution

Q105. AIPMT 2005

The correct order in which the O – O bond length increases in the following is

(A) $O_2 < H_2O_2 < O_3$ (B) $O_3 < H_2O_2 < O_2$

(C) $H_2O_2 < O_2 < O_3$ (D) $O_2 < O_3 < H_2O_2$

Open solution

Q106. AIPMT 2005

The surface tension of which of the following liquids is maximum ?

(A) C_2H_5OH (B) CH_3OH (C) H_2O (D) C_6H_6

Open solution

Q107. AIPMT 2005

The correct sequence of increasing covalent character is represented by

(A) $LiCl < NaCl < BeCl_2$ (B) $BeCl_2 < LiCl < NaCl$

(C) $NaCl < LiCl < BeCl_2$ (D) $BeCl_2 < NaCl < LiCl$

Open solution

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Q108. AIPMT 2005

Which of the following would have a permanent dipole moment?

(A) SiF_4 (B) SF_4 (C) XeF_4 (D) BF_3

Open solution

Q109. AIPMT 2004

Among the following, the pair in which the two species are not isostructural is

(A) SiF_4 and SF_4 (B) IO_3^- and XeO_3 (C) BH_4^- and NH_4^+ (D) PF_6^- and SF_6

Open solution

Q110. AIPMT 2004

In a regular octahedral molecule, MX_6 the number of $X - M - X$ bonds at 180° is

(A) three (B) two (C) six (D) four

Open solution

Q111. AIPMT 2004

H_2O is dipolar, whereas BeF_2 is not. It is because

(A) the electronegativity of F is greater than that of O

(B) H_2O involves hydrogen bonding whereas BeF_2 is a discrete molecule

(C) H_2O is linear and BeF_2 is angular

(D) H_2O is angular and BeF_2 is linear.

Open solution

Q112. AIPMT 2004

In BrF_3 molecule, the lone pairs occupy equatorial positions to minimize

(A) lone pair - bond pair repulsion only

(B) bond pair - bond pair repulsion only

(C) lone pair - lone pair repulsion and lone pair - bond pair repulsion

(D) lone pair - lone pair repulsion only.

Open solution

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Q113. AIPMT 2003

Which one of the following statements is not correct for sigma- and pi- bonds formed between two carbon atoms?

(A) sigma-bond is stronger than a pi-bond.

(B) Bond energies of sigma- and pi-bonds are of the order of 264 kJ/mol and 347 kJ/mol, respectively

(C) Free rotation of atoms about a sigma-bond is allowed but not in case of a pi-bond.

(D) Sigma-bond determines the direction between carbon atoms but a pi-bond has no primary effect in this regard.

Open solution

Q114. AIPMT 2002

Which of the following has $p\pi - d\pi$ bonding?

(A) NO_3^- (B) SO_3^{2-} (C) BO_3^{3-} (D) CO_3^{2-}

Open solution

Q115. AIPMT 2002

In NO_3^- ion number of bond pair and lone pair of electrons on nitrogen atom are

(A) 2, 2 (B) 3, 1 (C) 1, 3 (D) 4, 0

Open solution

Q116. AIPMT 2002

Which of the following is isoelectronic?

(A) CO_2, NO_2 (B) NO_2^-, CO_2 (C) CN^-, CO (D) SO_2, CO_2

Open solution

Q117. AIPMT 2001

Which of the following two are isostructural?

(A) XeF_2, IF_2^- (B) NH_3, BF_3 (C) CO_3^{2-}, SO_3^{2-} (D) PCl_5, ICl_5

Open solution

Q118. AIPMT 2001

In which of the following bond angle is maximum ?

(A) NH_3 (B) NH_4^+ (C) PCl_3 (D) SCl_2

Open solution

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Q119. AIPMT 2001

Nitrogen forms N_2, but phosphorus does not form P_2, however, it converts P_4, reason is

(A) triple bond present between phosphorus atom

(B) $p\pi - p\pi$ bonding is weak

(C) $p\pi - p\pi$ bonding is strong

(D) multiple bonds form easily.

Open solution

Q120. AIPMT 2001

In X - H --- Y, X and Y both are electronegative elements. Then

(A) electron density on X will increase and on H will decrease

(B) in both electron density will increase

(C) in both electron density will decrease

(D) on X electron density will decrease and on H increases.

Open solution

Q121. AIPMT 2000

$d\pi - p\pi$ bond present in

(A) CO_3^{2-} (B) PO_4^{3-} (C) NO_3^- (D) NO_2^-

Open solution

Q122. AIPMT 2000

Right order of dissociation energy N_2 and N_2^+ is

(A) $N_2 > N_2^+$ (B) $N_2 = N_2^+$ (C) $N_2^+ > N^2$ (D) none

Open solution

Q123. AIPMT 2000

Which species does not exhibit paramagnetism?

(A) N_2^+ (B) O_2^- (C) CO (D) NO

Open solution

Q124. AIPMT 2000

Which compound has planar structure?

(A) XeF_4 (B) $XeOF_2$ (C) XeO_2F_2 (D) XeO_4

Open solution

Q125. AIPMT 1998

The number of anti-bonding electrons in O_2^{2-} molecular ion on the basis of molecular orbital theory is (Atomic no. of O is 8)

(A) 5 (B) 2 (C) 4 (D) 6

Open solution

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Q126. AIPMT 1998

In PO_4^{3-} ion, the formal charge on each oxygen atom and P–O bond order respectively are

(A) $-0.75,\ 0.6$ (B) $-0.75,\ 1.0$ (C) $-0.75,\ 1.25$ (D) $-3,\ 1.25$

Open solution

Q127. AIPMT 1997

N_2 and O_2 are converted into monocations, N_2^+ and O_2^+ respectively. Which is wrong?

(A) In O_2 paramagnetism decreases. (B) N_2^+ becomes diamagnetic.

(C) In N_2, the N–N bond weakens. (D) In O_2, the O–O bond order increases.

Open solution

Q128. AIPMT 1997

N_2 and O_2 are converted into monoanions N_2^- and O_2^- respectively, which of the following statements is wrong?

(A) In O_2, bond length increases. (B) N_2^- becomes diamagnetic.

(C) In N_2, then N–N bond weakens. (D) In O_2, the O–O bond order increases.

Open solution

Q129. AIPMT 1997

The high density of water compared to ice is due to

(A) dipole-induced dipole interactions

(B) induced dipole induced dipole interactions

(C) hydrogen bonding interactions

(D) dipole-dipole interactions

Open solution

Q130. AIPMT 1996

The bond length between hybridised carbon atom and other carbon atom is minimum in

(A) propene (B) propyne (C) propane (D) butane.

Open solution

Q131. AIPMT 1996

Which of the following has sp^2-hybridisation?

(A) $BeCl_2$ (B) C_2H_2 (C) C_2H_6 (D) C_2H_4

Open solution

Q132. AIPMT 1996

The structure and hybridisation of $Si(CH_3)_4$ is

(A) octahedral, sp^3d (B) tetrahedral, sp^3 (C) bent, sp (D) trigonal, sp^2

Open solution

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Q133. AIPMT 1995

Which of the following species is paramagnetic?

(A) CO (B) CN^- (C) O_2^{2-} (D) NO

Open solution

Q134. AIPMT 1995

The correct order of the O–O bond length in O_2, H_2O_2 and O_3 is

(A) $O_2 > H_2O_2 > O_3$ (B) $H_2O_2 > O_3 > O_2$

(C) $O_2 > O_3 > H_2O_2$ (D) $O_3 > H_2O_2 > O_2$

Open solution

Q135. AIPMT 1995

The ground state electronic configuration of valence shell electrons in nitrogen molecule (N_2) is written as $KK, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi 2p_y^2 \sigma 2p_z^2$. Hence the bond order in nitrogen molecule is

(A) 2 (B) 3 (C) 0 (D) 1

Open solution

Q136. AIPMT 1995

In which of the following compounds there is more than one kind of hybridization (*sp*, sp^2, sp^3) for carbon?

(A) $CH_2{=}CH{-}CH{=}CH_2$ (B) $H{-}C{\equiv}C{-}H$

(C) $CH_3CH_2CH_2CH_3$ (D) $CH_3{-}CH{=}CH{-}CH_3$

Open solution

Q137. AIPMT 1995

The BCl_3 is a planar molecule whereas NCl_3 is pyramidal because

(A) nitrogen atom is smaller than boron atom

(B) BCl_3 has no lone pair but NCl_3 has a lone pair of electrons

(C) B–Cl bond is more polar than N–Cl bond

(D) N–Cl bond is more covalent than B–Cl bond.

Open solution

Q138. AIPMT 1994

Which of the following molecules has the highest bond order?

(A) O_2^- (B) O_2 (C) O_2^+ (D) O_2^{2-}

Open solution

Q139. AIPMT 1994

Which of the following molecule does not possess a permanent dipole moment?

(A) CS_2 (B) SO_3 (C) H_2S (D) SO_2

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Q140. AIPMT 1994

The table shown below gives the bond dissociation energies (E_{diss}) for single covalent bonds of carbon (C) atoms with element *A*, *B*, *C* and *D*. Which element has the smallest atoms?

Bond	E_{diss} $(kJ\ mol^{-1})$
C-*A*	240
C-*B*	328
C-*C*	276
C-*D*	485

(A) *C* (B) *D* (C) *A* (D) *B*

Open solution

Q141. AIPMT 1993

Among the following which compound will show the highest lattice energy?

(A) KF (B) NaF (C) CsF (D) RbF

Open solution

Q142. AIPMT 1993

Which one of the following is the correct order of interactions?

(A) Covalent $<$ hydrogen bonding $<$ van der Waals' $<$ dipole-dipole

(B) van der Waals' $<$ hydrogen bonding $<$ dipole $<$ covalent

(C) van der Waals' $<$ dipole-dipole $<$ hydrogen bonding $<$ covalent

(D) Dipole-dipole $<$ van der Waals' $<$ hydrogen bonding $<$ covalent.

Open solution

Q143. AIPMT 1993

Which of the following statements is not correct?

(A) Double bond is shorter than a single bond.

(B) Sigma bond is weaker than a π (pi) bond.

(C) Double bond is stronger than a single bond.

(D) Covalent bond is stronger than hydrogen bond.

Open solution

Q144. AIPMT 1993

When the hybridization state of carbon atom changes from sp^3 to sp^2 and finally to sp, the angle between the hybridized orbitals

(A) decreases gradually (B) decreases considerably

(C) is not affected (D) increases progressively.

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Q145. AIPMT 1993

Strongest bond is in between

(A) CsF (B) NaCl

(C) both (a) and (b) (D) none of the above.

Open solution

Q146. AIPMT 1993

Which of the following species has four lone pairs of electrons?

(A) I (B) O (C) Cl^- (D) He

Open solution

Q147. AIPMT 1992

Which one of the following has the shortest carbon carbon bond length?

(A) Benzene (B) Ethene (C) Ethyne (D) Ethane

Open solution

Q148. AIPMT 1992

Which structure is linear?

(A) SO_2 (B) CO_2 (C) CO_3^{2-} (D) SO_4^{2-}

Open solution

Q149. AIPMT 1992

Strongest hydrogen bond is shown by

(A) water (B) ammonia

(C) hydrogen fluoride (D) hydrogen sulphide.

Open solution

Q150. AIPMT 1992

Which of the following bonds will be most polar?

(A) $N-Cl$ (B) $O-F$ (C) $N-F$ (D) $N-N$

Open solution

Q151. AIPMT 1991

In compound *X*, all the bond angles are exactly $109^\circ 28'$, *X* is

(A) chloromethane (B) carbon tetrachloride

(C) iodoform (D) chloroform.

Open solution

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Q152. AIPMT 1991

The shortest C–C bond distance is found in

(A) diamond (B) ethane (C) benzene (D) acetylene.

Open solution

Q153. AIPMT 1991

A sp^3 hybrid orbital contains

(A) $1/4$ *s*-character (B) $1/2$ *s*-character (C) $1/3$ *s*-character (D) $2/3$ *s*-character.

Open solution

Q154. AIPMT 1991

A straight chain hydrocarbon has the molecular formula C_8H_{10}. The hybridization of the carbon atoms from one end of the chain to the other are respectively sp^3, sp^2, sp^2, sp^3, sp^2, sp^2, sp and sp. The structural formula of the hydrocarbon would be

(A) $CH_3C{\equiv}CCH_2{-}CH{=}CHCH{=}CH_2$ (B) $CH_3CH_2{-}CH{=}CHCH{=}CHC{\equiv}CH$

(C) $CH_3CH{=}CHCH_2{-}C{\equiv}CCH{=}CH_2$ (D) $CH_3CH{=}CHCH_2{-}CH{=}CHC{\equiv}CH$

Open solution

Q155. AIPMT 1990

Among $LiCl$, $BeCl_2$, BCl_3 and CCl_4, the covalent bond character follows the order

(A) $BeCl_2 > BCl_3 > CCl_4 < LiCl$ (B) $BeCl_2 < BCl_3 < CCl_4 < LiCl$

(C) $LiCl < BeCl_2 < BCl_3 < CCl_4$ (D) $LiCl > BeCl_2 > BCl_3 > CCl_4$

Open solution

Q156. AIPMT 1990

Which statement is NOT correct?

(A) A sigma bond is weaker than a pi bond.

(B) A sigma bond is stronger than a pi bond.

(C) A double bond is stronger than a single bond.

(D) A double bond is shorter than a single bond.

Open solution

Q157. AIPMT 1990

Which one shows maximum hydrogen bonding?

(A) H_2O (B) H_2Se (C) H_2S (D) HF

Open solution

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Q158. AIPMT 1990

Linear combination of two hybridized orbitals belonging to two atoms and each having one electron leads to the formation of

(A) sigma bond
(B) double bond
(C) co-ordinate covalent bond
(D) pi bond.

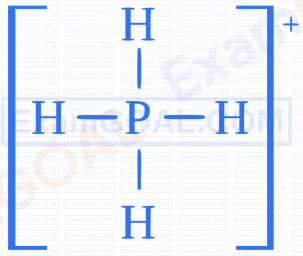
Open solution

Q159. AIPMT 1990

Which one of the following formulae does not correctly represent the bonding capacities of the two atoms involved?

Open solution

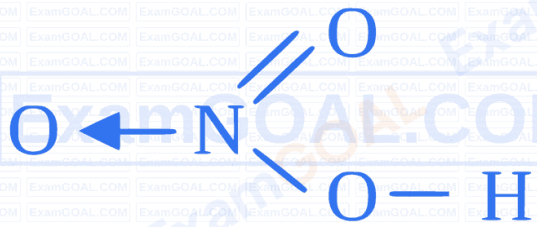
(A)



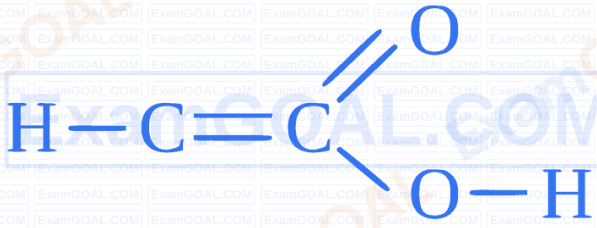
(B)



(C)



(D)



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Q160. AIPMT 1989

Which of the following molecule does not have a linear arrangement of atoms?

(A) H_2S (B) C_2H_2 (C) Be_2 (D) CO_2

Open solution

Q161. AIPMT 1989

In which one of the following molecules the central atom can be said to adopt sp^2 hybridization?

(A) BeF_2 (B) BF_3 (C) C_2H_2 (D) NH_3

Open solution

Q162. AIPMT 1989

H_2O has a net dipole moment while BeF_2 has zero dipole moment because

(A) H_2O molecule is linear while BeF_2 is bent

(B) BeF_2 molecule is linear while H_2O is bent

(C) fluorine has more electronegativity than oxygen

(D) beryllium has more electronegativity than oxygen.

Open solution

Q163. AIPMT 1989

Which of the following possesses a *sp*-carbon in its structure?

(A) $CH_2=CCl-CH=CH_2$ (B) $CCl_2=CCl_2$

(C) $CH_2=C=CH_2$ (D) $CH_2=CH-CH=CH_2$

Open solution

Q164. AIPMT 1988

The angle between the overlapping of one *s*-orbital and one *p*-orbital is

(A) 180° (B) 120° (C) $109^\circ 28'$ (D) $120^\circ, 60^\circ$

Open solution

Q165. AIPMT 1988

Equilateral shape has

(A) *sp* hybridisation (B) sp^2 hybridisation

(C) sp^3 hybridisation (D) dsp^3 hybridisation.

Open solution

Q166. AIPMT 1988

The Cl–C–Cl angle in 1, 1, 2, 2-tetrachloroethene and tetrachloromethane respectively will be about

(A) 120° and 109.5° (B) 90° and 109.5° (C) 109.5° and 90° (D) 109.5° and 120°

Open solution

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MCQ (More than One Correct Answer) 2 questions

Q167. NEET 2017

Which of the following pairs of compounds is isostructural ?

(A) TeI_2, XeF_2 (B) IBr_2^-, XeF_2 (C) IF_3, XeF_2 (D) $BeCl_2$, XeF_2

Open solution

Q168. NEET 2016 Phase 2

Which of the following pairs of ions is isoelectronic and isostructural ?

(A) CO_3^{2-}, NO_3^- (B) ClO_3^{2-}, CO_3^{2-} (C) SO_3^{2-}, NO_3^- (D) ClO_3^-, SO_3^{2-}

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **C**	3. **B**	4. **A**	5. **B**
6. **D**	7. **C**	8. **C**	9. **B**	10. **A**
11. **B**	12. **D**	13. **C**	14. **A**	15. **A**
16. **A**	17. **D**	18. **D**	19. **D**	20. **C**
21. **D**	22. **D**	23. **D**	24. **B**	25. **C**
26. **B**	27. **A**	28. **D**	29. **D**	30. **D**
31. **B**	32. **C**	33. **D**	34. **B**	35. **A**
36. **B**	37. **D**	38. **C**	39. **D**	40. **A**
41. **C**	42. **C**	43. **D**	44. **C**	45. **A**
46. **B**	47. **B**	48. **C**	49. **B**	50. **C**
51. **C**	52. **A**	53. **C**	54. **D**	55. **C**
56. **C**	57. **D**	58. **B**	59. **B**	60. **D**
61. **A**	62. **D**	63. **C**	64. **B**	65. **A**
66. **C**	67. **B**	68. **C**	69. **A**	70. **D**
71. **D**	72. **D**	73. **A**	74. **B**	75. **A**
76. **D**	77. **D**	78. **B**	79. **D**	80. **A**
81. **D**	82. **B**	83. **A**	84. **A**	85. **A**
86. **C**	87. **B**	88. **A**	89. **B**	90. **B**
91. **C**	92. **D**	93. **A**	94. **C**	95. **D**
96. **D**	97. **C**	98. **C**	99. **D**	100. **B**
101. **D**	102. **C**	103. **A**	104. **A**	105. **D**
106. **C**	107. **C**	108. **B**	109. **A**	110. **A**
111. **D**	112. **D**	113. **B**	114. **B**	115. **D**
116. **C**	117. **A**	118. **B**	119. **B**	120. **A**
121. **B**	122. **A**	123. **C**	124. **A**	125. **D**
126. **C**	127. **B**	128. **D**	129. **C**	130. **B**
131. **D**	132. **B**	133. **D**	134. **B**	135. **B**
136. **D**	137. **B**	138. **C**	139. **A**	140. **B**
141. **B**	142. **C**	143. **B**	144. **D**	145. **A**
146. **C**	147. **C**	148. **B**	149. **C**	150. **C**

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151. **B**	152. **D**	153. **A**	154. **D**	155. **C**
156. **A**	157. **A**	158. **A**	159. **D**	160. **A**
161. **B**	162. **B**	163. **C**	164. **A**	165. **B**
166. **A**				

MCQ (More than One Correct Answer)

167. **B, D**	168. **A**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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